

Optimizing parcel sorting process of vertical sorting system in e-commerce warehouse

Zheyi Tan, Haolin Li, Xueting He^{*}

School of Management, Shanghai University, Shanghai, China

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ABSTRACT

The vertical sorting system is an emerging system in warehouse section. It achieves higher level of automated sorting by applying conveyor sorter and AGVs. This paper investigates the optimization problem of parcel sorting in an e-commerce warehouse where parcels are waiting to be sorted and delivered. The assignment among the parcels, picking stations and AGVs are determined together with the objective of minimizing the finishing time of processing the last parcel, so that a detailed operational plan for warehouse sorting can be worked out. A mixed-integer linear programming model is developed based on the problem, and particle swarm optimization (PSO) algorithm is employed for problem solving. Numerical experiment shows that the proposed PSO algorithm can solve the model with high efficiency, and sensitivity analysis give suggestion about the arrangement of trailers' location.

1. Introduction

In the last decades, the rapid developments in business-to-consumer (B2C) e-commerce have led to disruptive growth in logistics section. A fast and accurate delivery at low cost is expected by both consumers and online retailers. Along with the expectation, warehouse operators are trying to improve warehouse performance to meet the requirement of small orders, large assortments, tight delivery schedules and varying workloads [1]. When dealing with the problem of “how to meet the logistics need”, some online retailers choose third-party logistics provider for shipping and delivery. However, several B2C retail giants such as Amazon and JD.com have made a different choice. They are building their own logistics system to maintain timely and personalized logistics service and improve customer experience. The private logistics system realizes the control of whole logistics chain, meanwhile, it raises a great challenge to online retailers on the control of the whole logistics process, especially when dealing with the complexity of warehousing activities like picking and sorting.

An efficiently managed e-commerce warehouse usually equips with smart warehouse systems for warehousing activities. For order picking process, automated picking systems such as robotic mobile fulfillment system (RMFS) and automated storage/retrieval system (AS/RS) are used to replace manual labor. The products picked by the automated systems are then packed into parcels according to orders, and sorted by

automated sorting system according to their destinations and wait for shipment. In the sorting process, automated sorting systems can sort a huge number of parcels with high speed and reliabilities. Among the different automated sorting systems, vertical sorting system is a relatively new one. It consists an induction station for loading and recognizing parcels, a sorting conveyor, several picking stations which are used to transfer sorted parcels to AGVs, and a fleet of AGVs used for handling parcels from picking stations to trailers. The automated picking stations and AGVs in vertical sorting system realize a higher degree of automation in the sorting and handling process of parcels.

When operating the highly-automated vertical sorting system, several decision problems need to be solved. First, we need to answer the question at which picking station will parcels be processed. Second, like other warehouse research with a fleet of AGVs, the assignment and scheduling of AGVs under some specific control policies need to be worked out. These decision problems are the basic problems for normal operation of vertical sorting system, at the same time, these problems could be complicated indeed. In this paper, we formulate a mixed-integer linear programming model for the assignment of vertical sorting system. Pooled strategy for AGV control is applied and modelled in our research, and the assignments of destinations, picking stations and AGVs are determined in the proposed model. To solve the problem in reasonable time, we propose a solution method with particle swarm optimization (PSO) algorithm.

^{*} Corresponding author.

E-mail address: hexueting@shu.edu.cn (X. He).

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The rest of the paper is organized as follows. The review of the state-of-art in automated order processing and sorting is presented in Section 2. The proposed problem is described in Section 3, while the proposed model for the model is presented in Section 4. The solution method is proposed in Section 5, and numerical experiment is presented in Section 6. At last, the conclusion is raised in Section 7.

2. Literature review

The vertical sorting system is relatively new in warehouse section. To our best knowledge, there are few works devoted to this field. Most works have focused on some other sortation systems which have similarities to vertical sorting system, like the conveyor-based sorting systems, robotic mobile fulfillment systems (RMFS), and some other AGV-based picking systems. Conveyor is an important part of vertical sorting system. Boysen et al. [2] have made a comprehensive review about automated sortation conveyors from operational research perspective. Their review focuses on the researches about layout design and system operation of conveyor-based systems. The research problems are divided into four categories, namely layout design, inbound problems, outbound problems and short-term scheduling decisions, which are used in literature classifications. Other systems, such as AGV-based picking systems and robot mobile fulfillment systems, are also widely used in automated order fulfillment [1,3]. Though these systems may differ in operating pattern and some other characteristics, they still can bring some insights to our research. The literature review in our research can be summed up into two streams, one is at the strategic level, the other is at the operational level.

The research on strategic level are mainly focus on some strategic decisions and plans such as system design and operation strategy. A typical conveyor-based system usually consists loading stations, conveyors and outbound docks. Fedtke and Boysen [4] investigates the impact of different design alternatives of tilt-tray sortation conveyor system on system throughput in a cross-docking center, including parallel conveyors, different number of loading stations and different loading rules. The solution procedures for the assignment of inbound and outbound destinations to docking doors is presented as a subordinate problem. van der Gaast et al. [5] study the impact of conveyor merges in sequential zone picking systems by building up an analytical model. An approximate model is built to study the impact on throughput capability by using an aggregation technique and matrix-geometric methods. As an extension of previous work, van der Gaast et al. [6] then develop a capacity model for sequential zone picking systems. Conveyor lanes, pick zones and other elements of the system are modelled as a queueing network with capacity constraints on sub-networks. An approximation model is proposed based on jump-over blocking. The two approximation models provide decision support tools for design of complex zone picking systems, in terms of number of segments and length of zones, input and output buffer capacities, and storage allocation of products to zones.

Some literature focuses on operational problems of conveyor system, which mainly investigate some specific operational decisions like destination assignment, scheduling and sequencing. Cross docks are special distribution centers in e-commerce supply chain and postal service which apply conveyors for shipment sorting and consolidation. Zenker and Boysen [7] extend the destination assignment problem of outbound docks to dock sharing problem, in which several destinations may share an outbound dock. A mixed-integer linear programming model is formulated where a partition of destinations is determined so that the maximum inventory in the system is minimized. A combined approach which combines greedy local search and tabu search is developed for problem solving. Briskorn et al. [8] test different decision rules of the assignment of parcels to trays in closed-loop tilt tray sortation conveyors, and formulate a short-term deterministic scheduling problem to minimize the makespan. Some insights on system configuration is also presented.

Robots and AGVs are the critical resources in RMFS and AGV-based picking systems, so the research on robot operational strategies have received much attention. The operational decision of RMFS and AGV-based picking system involves a wide range of decision problem, including rack and robot assignment, storage assignment, routing, scheduling and order sequencing. Among those problems, the robot assignment and scheduling problem is highly relevant to our research. Dehnavi-Arani et al. [9] propose an integrated model of conflict-free routing and job shop scheduling in an AGV-based system. Several AGVs are employed to transfer the jobs between machines and warehouse, and the job shop scheduling problem aims to find a proper assignment among the AGV fleet. The job assignment and the path selection of AGVs are both optimized so that the total make-span is minimized.

Another point we would like to address is the layout aspect of RMFS including the location of the workstations and the assignment of storage areas. Lamballais et al. [10] work out an evaluation model addressing layout aspects of RMFS. Queueing network models with high accuracy of robots driving behavior and multi-line orders are employed for both single-line and multi-line orders. The location of the workstations can influence maximum order throughput, while the effect of changing length-to-width ratio of storage area is limited to order throughput. Wang, Yang et al. [11] investigate the operation cycle of rack-moving robot from the perspective of the lane depth, and proposes expected travel time models of the rack-moving robot for single- and multi-deep layouts warehouses. Lee and Murray [12] define pick, place, and transport vehicle routing problem in robotic warehouse that minimize the time to deliver all items from a pick list to the packing station. A mixed-integer linear programming model is developed to investigate the influence of different layouts including cross aisle, location of packing station and average travelling distance on warehouse performance. Boysen, Briskorn et al. [13] investigate the optimization of order processing problem in Kiva system where rack visits and order sequence are optimized. They also investigated the impact of shared storage policy on the number of the robot fleet.

A common assumption in previous research and practice is that every robot is dedicated to a specific picking station, which is often referred as the dedicated strategy. When applying robots pooled strategy, robots can serve all the stations as the need of all the picking stations, and the system is operated in a centralized manner. Yuan and Gong [14] apply pooled strategy into RMFS, and build open queueing network models to describe RMFS with two protocols in sharing robots for order pickers. The advantage of robots pooled strategy is presented by the reduction of optimal number of robots needed and the throughput time in RMFS. He et al. [15] develop a differentiated service policy in AGV system with differentiated probabilistic queueing strategy for order scheduling with different classes. Each order is probabilistically assigned to an AGV in the proposed strategy, while some AGVs are reserved only for processing certain classes of orders as a special case. Roy et al. [16] use analytical models to compare the efficiency of dedicated and pooled zone policy in the AGV picking system with the storage settings of both single and multiple storage zones. Multi-class closed queueing network models are applied for the comparison of the two strategies. Pooled strategy outperforms dedicated strategy in terms of expected throughput time of picking process, however, the replenishment process under pooled robot strategy may take longer time. The performance of different robot zoning policies are evaluated by a two-stage stochastic model, where shortest queue zone policy and dedicated zone policy result in similar system performance, and random zone assignment policy underperforms other policies.

To obtain feasible assignments of both picking stations and robots, some heuristic algorithms are employed. Some meta-heuristics are applied to accelerate convergence, like Genetic algorithm [17,18], Squeaky Wheel Optimization [19], Critical-Shaking Neighborhood Search [20,21], and column generation-based heuristic [22]. Introduced by Kennedy and Eberhart [23] by the inspiration of bird flocking, PSO is

Table 1
Summary of reviewed literature.

Article Number	System	Problem	Model	Approach
[4]	C	DE	IP	Greedy randomized adaptive search, simulated annealing
[8]	C	SC	AM	Exact and polynomial algorithms
[5]	C	DE	AM, QN	Approximation model
[6]	C	DE	QN	Approximation model
[10]	R	DE	AM, QN	Simulation validation
[15]	R	OS	QN	Alternating minimization algorithm, Simulated Annealing algorithm
[14]	R	OS	QN	Simulation validation
[16]	R	OS	AM, QN	Simulation validation
[7]	C	AS	IP	Greedy local search, tabu search
[9]	R	SC	IP	Ant colony algorithm
[26]	R	SC	IP	Adaptive large neighborhood search heuristic is developed
This paper	C+R	AS	IP	PSO algorithm

Notes: C: conveyor-based systems. R: RMFS and AGV-based systems. O: other picking systems. DE: design. SC: sequencing and scheduling. OS: operation strategy, AS: order picking and sorting assignment. IP: integer programming. AM: analytical model. QN: queuing network.

a promising and widely-used heuristic in optimization and artificial intelligence. In recent years, PSO shows great capability in solving logistics optimization problems, for example, yard allocation problems in container terminals [24,25].

A brief summary of reviewed literature is shown in Table 1. From the reviewed literatures, we can find that though many works have been done to the strategic and operational decision of automated sorting and order processing, to our best knowledge, there are few forks that aim to the operational decision of vertical sorting system. The main contribution of our research can be categorized into three aspects. First, we investigate the operation problem of vertical sorting system, and formulate the problem as a mixed-integer programming model. Second, we proposed PSO algorithm for problem solving which provides a

method to solve real-scale problem. Third, some managerial insights about location assignment of trailers are concluded according to numerical experiment and sensitivity analysis, which provides a broad view for system operation.

3. Problem description

Consider the sorting process of packed parcel in an e-commerce warehouse equipped with vertical sorting system. Parcels are picked and packed by previous process, and waiting to be sorted by vertical sorting system before delivery. The main components of a vertical sorting system include a cross-belt conveyor with an induction station and several picking stations and a fleet of AGVs. The cross-belt conveyor is used to sort parcels. The induction station is used to feed parcels to the system and recognize destination information of parcels which would be used in sorting. A picking station has a chute for receiving parcels from the cross-belt conveyor and a mechanical arm for putting parcels onto AGVs. The AGV fleet is used to handling parcels from picking stations to trailers. In our research, the pooled strategy for AGV control is investigated, in which AGVs can assign to serve all the picking stations as the need of parcels. Those AGVs in idle state are parked in the parking area, and go to the assigned picking stations when received handling tasks.

Based on the system configuration mentioned above, a typical parcel sorting process contains the following steps.

Step 1: Parcels are released into the system sequentially, and are loaded onto the conveyors by induction station. At the same time, the induction station recognizes parcels' information for the incoming sorting process.

Step 2: Parcels travel along the conveyors and are sorted by the cross-belt conveyor according to the parcel's destinations. When a parcel reaches its assigned picking station, the cross-belt leads the parcel slides onto the chute and wait to be processed by the picking station.

Step 3: Parcels are handled by the picking stations and AGVs. AGVs move to assigned picking stations according to the assignment. Once the assigned AGV arrives the picking station, a parcel is loaded onto AGV by the robot arm of picking station.

Step 4: The AGV loaded with parcel moves to the position of assigned trailer, and drop the parcel into the trailer. Then the AGV goes to the

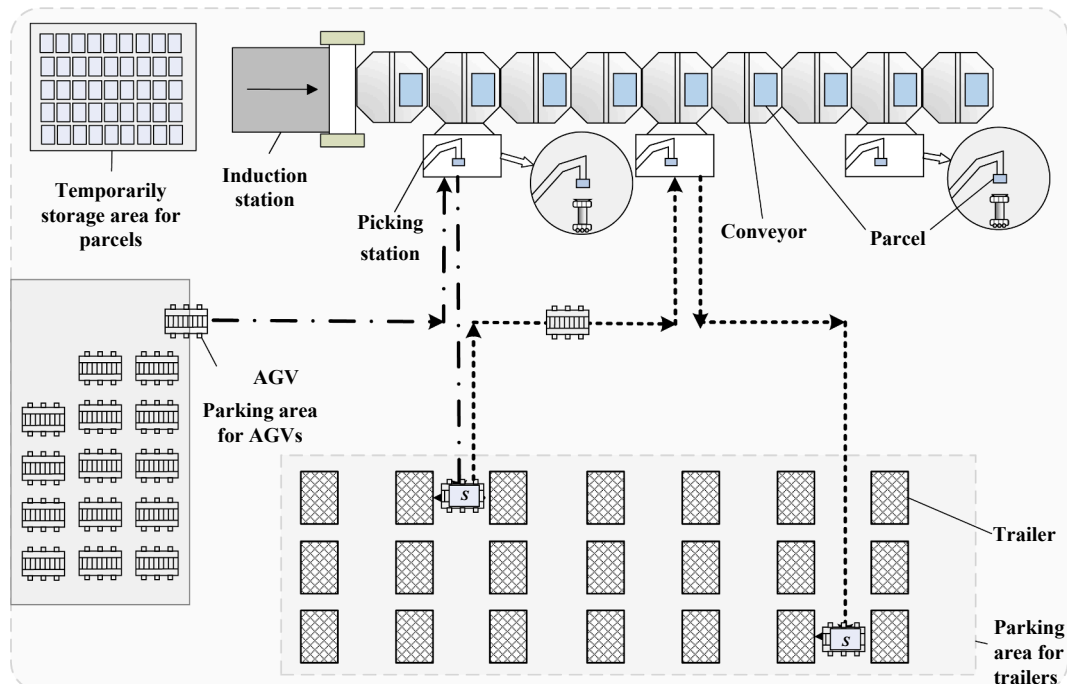


Fig. 1. A top view of vertical sorting system and the sorting process.

assigned picking station to wait for the next parcel.

Fig. 1 depicts an example of the sorting process in the vertical sorting system. From the above steps, we can extract several critical decision problems. In step 2, we need to assign parcels to picking stations according to different destinations, which is a common decision problem of sorting system. The step 3 and 4 require a detailed plan of AGVs including task assignment and scheduling. The e-commerce logistics activities are time-sensitive, so time is a critical factor in improving warehouse efficiency and customer experience. In our research, the finish time of the last parcel is selected as the optimization objective.

Before modelling the sorting problem, several assumptions need to be clarified.

- A picking station can only process parcels for a certain destination, and parcels to the same destination can only be proceeded by a certain picking station. For a further step, there are enough picking stations that can satisfy the need of all the destinations. This assumption is carried out for the purpose of reducing process complexity of picking stations, which is in line with the research of Fedtke and Boysen [4], and can be found in warehouse practice.
- Parcels to be released to the system are packed and waited to be sorted in a temporary storage area. The destination information of parcels and the sorting sequence of parcels for a certain destination are known in advance. Further, we assume that all the parcels are standard-sized.

4. Mathematical model

In this section, we formulate a mixed-integer programming model for the sorting problem. The objective of this model is to minimize the completion time of the last parcel of all the parcels. The assignment of parcels, picking stations and AGVs, and the schedule of AGVs is determined simultaneously in the model.

4.1. Notations

The following notations are introduced to the model.

Indices and sets	
S	Set of parcel numbers.
S_1	A subset of S , where $S_1 \subseteq S, 2 \leq S_1 \leq S - 1$.
$e(r), e(r')$	Dummy points for the start point and end point of AGV routes. $S^0 = S \cup e(r), \bar{S} = S \cup e(r'), \hat{S} = S \cup e(r) \cup e(r')$
N	Set of picking stations.
R	Set of AGVs.
D	Set of parcel destinations.
Parameters	
$a_{s,s'}$	Parcel release sequence index, set to 1 when parcel s and s' have the same destination, and parcel s' is released to the system right after parcel s .
b_s	Destination of parcel s .
$c_{r,n}^1$	Travel time of AGV r from parking area to picking station n .
$c_{n,s}^2$	AGV travel time from picking station n to trailer for parcel s .
c_n^3	Parcel travel time of conveyor from induction station to picking station n .
g_s	The time of parcel s released to the system.
t	The handling time of mechanical arm
Decision variables	
$\alpha_{n,d}$	Binary, set to 1 if the parcels to destination d will be processed by workstation n .
$\beta_{s,s',r}$	Binary, set to 1 if parcel s' is handled by AGV r right after parcel s .
$\varepsilon_{s,r}$	Binary, set to 1 if parcel s is handled by AGV r .
$\gamma_{s,r}$	Float, the arrival time of AGV r to the workstation which processes parcel s .
$\delta_{s,r}$	Float, the start time of parcel s being handled by AGV r .
ω_r	Float, the time of AGV r finishing all its sorting tasks.

4.2. Mathematical model

$$\text{Minimize } \max_{r \in R} \{\omega_r\} \quad (1)$$

$$\sum_{n \in N} \alpha_{n,d} = 1 \quad \forall d \in D \quad (2)$$

$$\sum_{d \in D} \alpha_{n,d} \leq 1 \quad \forall n \in N \quad (3)$$

$$\sum_{s \in S^0} \beta_{s,s',r} = \sum_{s \in \bar{S}} \beta_{s',s,r} = \varepsilon_{s',r} \quad \forall s' \in S, r \in R \quad (4)$$

$$\sum_{s \in \bar{S}} \beta_{e(r),s,r} = \sum_{s \in S^0} \beta_{s,e(r'),r} = 1 \quad \forall r \in R \quad (5)$$

$$\sum_{s \in S_1} \sum_{s' \in S_1} \beta_{s,s',r} \leq |S_1| \quad \forall S_1 \subseteq S, 2 \leq |S_1| \leq |S| - 1, \forall r \in R \quad (6)$$

$$\sum_{r \in R} \sum_{s \in S^0} \beta_{s,s',r} = \sum_{r \in R} \sum_{s \in \bar{S}} \beta_{s',s,r} = 1 \quad \forall s' \in S \quad (7)$$

$$\gamma_{s,r} \geq \sum_{n \in N} c_{r,n}^1 \alpha_{n,b_s} + M(\beta_{e(r),s,r} - 1) \quad \forall s \in S, r \in R \quad (8)$$

$$\gamma_{s',r} \leq \delta_{s,r} + t + \sum_{n \in N} c_{n,s}^2 \alpha_{n,b_s} + \sum_{n \in N} c_{n,s'}^2 \alpha_{n,b_{s'}} + M(1 - \beta_{s,s',r}) \quad \forall s, s' \in S, r \in R \quad (9)$$

$$\gamma_{s',r} \geq \delta_{s,r} + t + \sum_{n \in N} c_{n,s}^2 \alpha_{n,b_s} + \sum_{n \in N} c_{n,s'}^2 \alpha_{n,b_{s'}} - M(1 - \beta_{s,s',r}) \quad \forall s, s' \in S, r \in R \quad (10)$$

$$\delta_{s',r} \geq \delta_{s,r} + t - M \left(3 - a_{s,s'} - \sum_{s_1 \in S^0} \beta_{s_1,s,r} - \sum_{s_1 \in \bar{S}} \beta_{s',s_1,r} \right) \quad \forall s, s' \in S, r \in R \quad (11)$$

$$\gamma_{s,r} \leq \delta_{s,r} \quad \forall s \in \hat{S}, r \in R \quad (12)$$

$$\delta_{s,r} \leq M \varepsilon_{s,r} \quad \forall s \in \hat{S}, r \in R \quad (13)$$

$$\delta_{s,r} \geq g_s + \sum_{n \in N} c_{n,s}^3 \alpha_{n,b_s} + M(\varepsilon_{s,r} - 1) \quad \forall s \in S, r \in R \quad (14)$$

$$\omega_r \geq \delta_{s,r} + t + \sum_{n \in N} c_{n,s}^2 \alpha_{n,b_s} \quad \forall s \in S, r \in R \quad (15)$$

$$\alpha_{n,d}, \beta_{s,s',r}, \varepsilon_r \in \{0, 1\} \quad \forall s, s' \in \hat{S}, r \in R, d \in D \quad (16)$$

$$\gamma_{s,r}, \delta_{s,r} \geq 0 \quad \forall s \in \hat{S}, r \in R \quad (17)$$

The objective of this model is to minimize the finishing time of processing the last parcel. Constraints (2) and (3) state the corresponding relationship between the parcel destinations and workstations, that is, parcels for the same destination will be processed by a dedicated workstation, and a workstation can only process parcels for only one destination. Constraints (4) and (5) ensure the consecutiveness of the tasks assigned to each AGVs, and constraints (6) prevent the AGV paths forming loops. Constraints (7) make sure that a parcel must be handled by one AGV. Constraints (8) express that the arrival time of the AGV handling parcel s cannot be earlier than the arriving time from parking area to the picking station if parcel s is the first parcel being processed for its destination. Constraints (9) and (10) specify that if parcel s' is processed by AGV r right after parcel s , the arrival time of the AGV handling parcel s' should be later than the time that AGV goes back to the picking station after finishing the handling process of parcel s . Constraints (11) specify that if parcel s' is processed right after parcel s

by the same picking station, the start time of parcel s being handled by AGV r needs to be later than the time parcel s being handled by the picking station, that is, the time that parcels being handled by AGV plus the handling time of the mechanical arm. Constraints (12) make sure that the start time of an AGV handling parcel must be later than the time that the AGV arrives the picking station. Constraints (13) make sure that the AGV will arrive the picking station to process a parcel only when the parcel is assigned to be handled by the AGV. Constraints (14) make sure that the start time of an AGV processing a parcel need to be later than the time that the parcel arrives the picking station. Constraints (15) state the finishing time of processing the last parcel Constraints (16) and (17) define the decision variables.

5. Solution methodology

Like other optimization problem, for some small instances, the proposed model can be solved by CPLEX and some other commercial solver in a short time. However, when facing larger instances, commercial solver can hardly solve the problem in reasonable time. To deal with the problem, we introduce PSO algorithm for problem solving.

5.1. Solution representation and velocity updating strategy

In the population of PSO, each solution is denoted by a particle containing the statues of its position and velocity at each iteration. The particle's position denotes the solution quality corresponding with the particle, and the particle's velocity denotes the direction which it will move in the next iteration.

From the constraints (4) of the proposed model, the decision variables $\beta_{s,s,r}$ and $\varepsilon_{s,r}$ are related to each other. Meanwhile, from the constraints (8)–(14), the decision variables $\gamma_{s,r}, \delta_{s,r}$ can be derived by $\alpha_{n,d}, \beta_{s,s,r}, \varepsilon_{s,r}$. When knowing the value of variables $\alpha_{n,d}, \beta_{s,s,r}, \varepsilon_{s,r}$, the variables $\gamma_{s,r}, \delta_{s,r}$ can be determined with the help of CPLEX. Based on these characteristics, we design a PSO method to decide the variables $\alpha_{n,d}, \beta_{s,s,r}, \varepsilon_{s,r}$, and then using CPLEX to derive the value of $\gamma_{s,r}, \delta_{s,r}$ for further iterations.

For the PSO method, we define two groups of particle positions. To improve solving efficiency, we apply a dimensional reduction method for the particles. The first group of particles is defined by variables $\alpha_{n,d}$ with the notation $P^1_{g,m,d}$. For the particle m in iteration g , $P^1_{g,m,d}$ represents the number of picking station which is assigned to destination d . Similarly, the second group of particles is defined by variables $\varepsilon_{s,r}$ with the notation $P^2_{g,m,s}$. For the particle m in iteration g , $P^2_{g,m,s}$ represents the number of AGV which is assigned to parcel s . The updating formula of velocity and position for both groups of particles are presented as follows:

$$V^1_{(g+1),m,d} = V^1_{g,m,d} + c_1 r_1 (pbest^1_{g,m,d} - P^1_{g,m,d}) + c_2 r_2 (gbest^1_{g,m,d} - P^1_{g,m,d}) \quad (18)$$

$$P^1_{(g+1),m,d} = P^1_{g,m,d} + V^1_{(g+1),m,d} \quad (19)$$

$$V^2_{(g+1),m,s} = V^2_{g,m,s} + c_1 r_1 (pbest^2_{g,m,s} - P^2_{g,m,s}) + c_2 r_2 (gbest^2_{g,m,s} - P^2_{g,m,s}) \quad (20)$$

$$P^2_{(g+1),m,s} = P^2_{g,m,s} + V^2_{(g+1),m,s} \quad (21)$$

In formula (18) and (19), $V^1_{(g+1),m,d}$ and $V^1_{g,m,d}$ represent the current velocity and the previous velocity of particle $P^1_{g,m,d}$ on dimension m , respectively. For dimension m , $pbest^1_{g,m,d}$ denotes the best position of particle $P^1_{g,m,d}$ on up to iteration g , and $gbest^1_{g,m,d}$ denotes the best position of the whole swarm until iteration g ; $P^1_{(g+1),m,d}$ and $P^1_{g,m,d}$ denote the current and previous position of the particle. In formula (20) and (21),

Table 2
PSO algorithm procedure.

Set: M	// M is the set of particles.
Parameters: $g, G, M , c_1, c_2, r_1, r_2$	// g represents the current iteration number, G represents the maximum iteration number, $ M $ represents the population size, c_1, c_2 represent the acceleration weights, r_1, r_2 represent the learning factors
Objective: ω_r	// The finish time of all the sorting tasks.
1	Define $GBest_g, PBest_{g,m}, P^1_{g,m,d}, V^1_{g,m,d}, P^2_{g,m,s}, V^2_{g,m,s}, pbest^1_{g,m,d}, gbest^1_{g,m,d}, pbest^2_{g,m,s}, gbest^2_{g,m,s}, Fitness_{g,m}$
2	For $m \in M$
3	For $d \in D$
4	Set $P^1_{g,m,d}$ to a random number of intervals (0, N)
5	End for
6	For $s \in S$
7	Set $P^2_{g,m,s}$ to a random number of intervals (0, R)
8	End for
9	Obtain the value of $\beta_{s,s,r}$ by $P^2_{g,m,s}$ and $a_{s,s}$
10	Substitute $P^1_{g,m,d}, P^2_{g,m,s}$, and $\beta_{s,s,r}$ into CPLEX model and obtain the value of $\gamma_{s,r}, \delta_{s,r}, \omega_r$ and the corresponding fitness value.
11	Update $PBest_{g,m}, GBest_g$
12	End for
13	While ($g < G$) do
14	For $m \in M$
15	For $d \in D, s \in S$
16	Update $V^1_{g,m,d}, P^1_{g,m,d}, V^2_{g,m,s}, P^2_{g,m,s}$
17	If $P^1_{g,m,d} < 0$, then $P^1_{g,m,d} = 0$; If $P^1_{g,m,d} > N$, then $P^1_{g,m,d} = N$
	If $P^2_{g,m,s} < 0$, then $P^2_{g,m,s} = 0$; If $P^2_{g,m,s} > R$, then $P^2_{g,m,s} = R$
18	End for
19	Substitute $P^1_{g,m,d}, P^2_{g,m,s}$, and $\beta_{s,s,r}$ into CPLEX model and obtain the value of $\gamma_{s,r}, \delta_{s,r}, \omega_r$ and the corresponding fitness value.
20	Update $PBest_{g,m}, GBest_g$
21	End for
22	$g = g + 1$
23	End while
24	Return the objective value

$V^2_{(g+1),m,s}$ and $V^2_{g,m,s}$ represent the current velocity and the previous velocity of particle $P^2_{g,m,s}$ on dimension m , respectively. For dimension m , $pbest^2_{g,m,s}$ denotes the best position of particle $P^2_{g,m,s}$ on up to iteration g , and $gbest^2_{g,m,s}$ denotes the best position of the whole swarm until iteration g ; $P^2_{(g+1),m,s}$ and $P^2_{g,m,s}$ denote the current and previous position of the particle. For both groups of particles, c_1 and c_2 are acceleration weights. r_1 and r_2 are learning factors randomly generated within the interval [0, 1].

Since we apply dimensional reduction in solution representation, the continuous position of particles should be changed into the binary form of decision variables. The modification is listed as follows:

$$\alpha_{n,d} = \begin{cases} 1, & \text{when } n \leq P^1_{g,m,d} < n+1 \\ 0, & \text{otherwise} \end{cases} \quad (22)$$

$$\varepsilon_{s,r} = \begin{cases} 1, & \text{when } r \leq P^2_{g,m,s} < r+1 \\ 0, & \text{otherwise} \end{cases} \quad (23)$$

5.2. Main framework of the PSO procedure

Based on the above components, the PSO procedure for the proposed problem is listed in Table 2.

6. Numerical experiment

To validate the effectiveness of the proposed model and efficiency of the proposed algorithm, we conduct extensive numerical experiments on different instances. The experiment is conducted on a workstation

Table 3
Algorithm performance for small-scale instances.

Instances		CPLEX		Heuristic	PSO			
Scale	ID	F_{CPLEX}	t_{CPLEX}	F_{heu}	F_{PSO}	t_{PSO}	GAP_c	GAP_g
$ S = 8$	1	59	8.07	74	59	46.61	0.00%	25.42%
	2	54	5.59	65	54	19.38	0.00%	20.37%
	3	65	5.22	80	65	25.69	0.00%	23.08%
	4	69	10.39	84	69	21.99	0.00%	21.74%
	5	55	7.13	70	55	25.58	0.00%	27.27%
$ S = 10$	1	70	223.37	80	70	65.49	0.00%	14.29%
	2	60	192.47	70	60	40.58	0.00%	16.67%
	3	58	255.08	80	58	40.57	0.00%	37.93%
	4	67	98.89	76	68	14.88	1.49%	11.76%
	5	81	443.73	101	81	30.56	0.00%	24.69%
$ S = 12$	1	70	>7200	91	70	21.83	0.00%	30.00%
	2	91	>7200	135	89	40.03	-2.20%	51.69%
	3	97	>7200	115	93	41.72	-4.12%	23.66%
	4	92	>7200	110	89	37.59	-3.26%	23.60%
	5	84	>7200	93	82	36.79	-2.38%	13.41%

Notes: (1) F_{CPLEX} denotes the optimal solution obtained by CPLEX. For the instance scale $|N| = |D| = 3$, $|R| = 4$, $|S| = 12$, F_{CPLEX} denotes the feasible solution obtained by CPLEX in 7200 s. F_{heu} denotes the feasible solution obtained by the heuristic. F_{PSO} denotes the global best solution obtained by PSO algorithm. (2) t_{CPLEX} is the computation time of CPLEX in seconds. t_{PSO} is the computation time of PSO algorithm in seconds. (3) $GAP_c = (F_{PSO} - F_{CPLEX})/F_{CPLEX}$, $GAP_g = (F_{heu} - F_{PSO})/F_{PSO}$.

Table 4
Algorithm performance for large-scale instances.

Instances		Heuristic	PSO		
Scale	ID	F_{heu}	F_{PSO}	t_{PSO}	GAP_g
$ S = 20$	1	136	105	168.72	29.52%
	2	124	106	148.86	16.98%
	3	136	100	132.58	36.00%
	4	120	97	167.97	23.71%
	5	163	132	180.12	23.48%
$ S = 40$	1	275	218	361.35	26.15%
	2	264	214	140.56	23.36%
	3	256	219	140.54	16.89%
	4	300	216	363.79	38.89%
	5	271	208	365.07	30.29%
$ S = 80$	1	469	406	818.10	15.52%
	2	539	455	801.21	18.46%
	3	585	470	298.78	24.47%
	4	565	480	334.70	17.71%
	5	555	477	178.7	16.35%

Note: $GAP_g = (F_{heu} - F_{PSO})/F_{PSO}$.

equipped with Intel i5-10210U CPU @1.60 GHz and 16 GB RAM. The proposed model and algorithm are implemented in C# (VS2019) concert technology with the solver IBM ILOG CPLEX 12.6.1.

6.1. Parameter setting

The parameters of PSO algorithm are set as follows: the maximum number of iterations (G) is 100, the population size ($|M|$) is 100. The acceleration weights c_1 and c_2 are both set to 0.638. The learning factors r_1 and r_2 are randomly generated within the interval $[0, 1]$. For the input parameter of instances, the small-scale instances are all generated with the scale $|N| = |D| = 3$, $|R| = 4$, and the large-scale instances are all generated with the scale $|N| = |D| = 4$, $|R| = 6$. The destination of parcels b_s is generated randomly. The release time of parcels is related to the operation of induction station, and we use the formula $g_s = \frac{8}{27}s$ to generate the release time. The parameters of AGV travel time $c_{r,n}^1$, $c_{n,s}^2$ and c_n^3 are predetermined according to the layout of the system.

6.2. Algorithm performance

We conduct extensive experiments to assess the solution quality and

efficiency of PSO algorithm. The results are shown in Tables 3 and 4, with the objective values obtained by different methods, computation time and the gap listed. An upper limit of 7200 s is set as the maximum computing time. In our experiment, a heuristic algorithm is carried out for comparison. The procedure of the heuristic is listed as follows.

Step 1: Assigning destinations to picking stations according to the order of number, for example, the destination 1 is assigned to the picking station 1, the destination 2 is assigned to picking station 2, and so on.

Step 2: Assign AGVs to destinations (picking stations) according to the proportion of parcels. In this step, each AGV is dedicated to a specific destination. For example, if there are 10 AGVs numbered 1–10, and the parcels to destination 1 account for 20% of the total, parcels to destination 2 accounts 30%, then AGV 1 and 2 will be assigned to destination 1, AGV 3, 4 and 5 will be assigned to destination 2.

Step 3: When more than one AGV is assigned to a destination (picking station), the sequence of AGVs handling packages will be in a cycle. Based on the previous example, the sequence of AGVs in picking station 1 is 1-2-1-2-...

Experiments on small-scale instances are conducted first by comparing the solution obtained by CPLEX, PSO algorithm and a heuristic. From Table 3, we can see that CPLEX can only solve some small-scale problem. When the amount of parcels increase from 8 to 10, the computation time of CPLEX significantly increase. CPLEX cannot solve the instances with 12 parcels in reasonable time. PSO algorithm results in high solution efficiency as the gap between the solution obtained by CPLEX and PSO algorithm close to 0. What's more, the computation time of PSO algorithm is more stable when the scale of instances increases, where the average computing time for PSO is 33.95 s. For the instance scale $|N| = |D| = 3$, $|R| = 4$, $|S| = 10$ and $|N| = |D| = 3$, $|R| = 4$, $|S| = 12$, PSO algorithm result in much shorter computation time than CPLEX. The solution obtained by PSO is significantly better than the solution obtained by the heuristic. The above result shows that PSO algorithm is capable for small-scale instances.

For experiments of large-scale instances, we compare the result obtained by PSO algorithm and the heuristic. The result in Table 4 shows that PSO algorithm is competent to solve the large-scale problem, with the computing time ranging from 132.58 to 818.10 s, and 306.74 s on average. From the gap we can see that the solution obtained by PSO algorithm is much better than that obtained by the heuristic. The average gap between the two solutions is 23.85%, that's to say, compared with the average number 24.37% of GAP_g in small-scale instances, PSO algorithm results in good performance in solving large-scale problem.

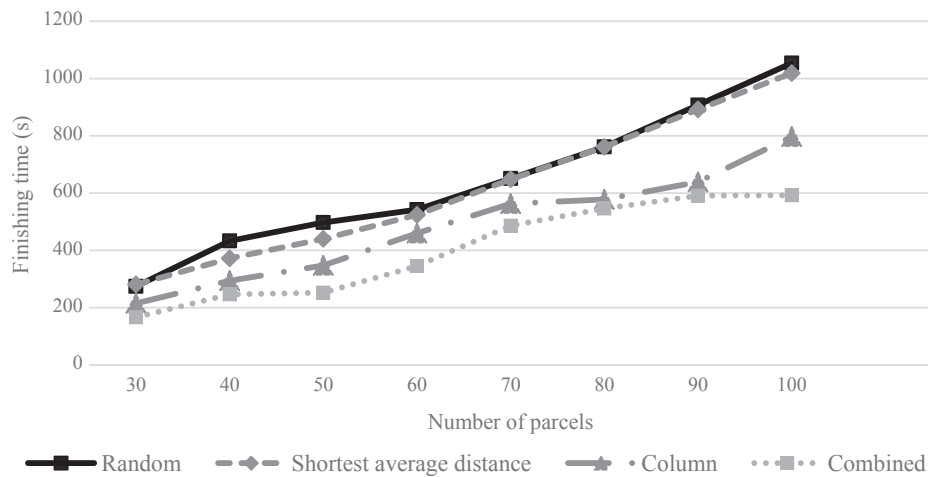


Fig. 2. The result of the sensitivity analysis about different trailer layouts.

6.3. Sensitivity analysis

In the proposed system, the total finish time is closely connected to the travel time of AGVs, where the travel distance can greatly influence AGV's travel time. In fact, the location of trailers can significantly influence the performance. The misconfiguration on trailers' location will lead to more useless AGV travel distance in the handling process. To discover the influence of different arrangements of trailer locations, we conduct a sensitivity analysis in the numerical experiments. There are four different arrangements examined by experiments.

Arrangement 1: Random. Trailers for different destinations are randomly distributed among the locations of the trailer parking area.

Arrangement 2: Shortest average distance. The locations of trailers are determined by the number of packages to be assigned. The trailers with more parcels to be loaded are assigned to the locations with a lower average distance to picking stations.

Arrangement 3: Column. The trailers which load packages to the same destination will be arranged in the same column. In this layout, the trailers and the corresponding picking station are positioned at the column as close as possible to the corresponding picking station.

Arrangement 4: Combined. Based on the column pattern layout, for the trailers in the same row, the trailer with more parcels to be loaded will be assigned to the location nearer to picking stations.

The results of different layouts are compared in Fig. 2. From the figure we can conclude that the random layout results the worst performance among the four different layouts. This is because that there may be some misconfiguration in random layout. The finishing time of shortest average distance layout is slightly lower than the result of random layout, while the column arrangement significantly reduces the finishing time since it makes the further improvement in reducing AGVs' travel distance. The combined layout performs best among the four layouts. From this point, warehouse operator could use the combined arrangement for further improvement of warehouse efficiency.

7. Conclusion

This paper investigates the sorting problem of vertical sorting system. In the vertical sorting system, the sorting conveyor, and the AGVs which are controlled by pooled strategy are used to handle sorted parcels. A mixed-integer linear programming model is formulated which aims to minimize total sorting time and determines the assignments of destinations, picking stations and AGVs. We apply the PSO algorithm for problem solving. The major contributions are summarized in the following aspects:

1. Our paper extends the research scope of warehouse operation, especially automated sorting. The vertical sorting system is a relatively new system for parcel sorting in the e-commerce warehouse. In our research, the sorting assignment of vertical sorting system is investigated, which provides a decision-making tool for the operation of vertical sorting system.
2. PSO algorithm is applied in problem solving, which is competent in solving both small and large-scale problems. PSO algorithm improves computing efficiency, so the research finding provides a decision support tool for warehouse operation.
3. Different warehouse layout will lead to different operational efficiency. The arrangement combining both shortest average distance arrangement and column arrangement for trailer's location outperforms other arrangements according to the result of sensitivity analysis, so warehouse operators could use the combined arrangement for further improvement of warehouse efficiency.

This research also has its limitations. In our research, we only investigate the circumstance that only one destination is assigned to the picking station. In some cases, a picking station can be assigned with several destinations, which can increase the utilization of some picking stations. For robot control strategy, we only investigate pooled strategy in our research, while dedicated strategy may bring some other insights in system operation. Future research can work on the application of different operation strategy and some new extensions of the proposed system.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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